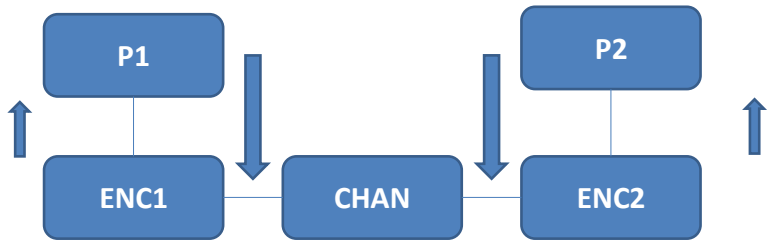
University of Athens - Department of Informatics and Telecommunications

Operating Systems (K22) / Period 2020-2021 / 1st Assignment

(Instructor: E.Hadjiefthymiades)

Implement a two-layer message exchange architecture according to the figure below. There are five processes P1, P2, ENC1, ENC2 and CHAN. Processes P{1|2} accept and present data to the user. The input to processes P{1|2} is a line of text which is registered as a message with the prospect of being sent to the other process (that is P1 → P2 or P2 → P1). If the message is registered in P1 it will be "transmitted" by the underlying layers to P2 and if the correctness of the message is verified it will be presented to the user (of P2). Processes ENC1 and ENC2 accept messages from P{1|2}, compute the hash value of the given string (message checksum) and then forward the augmented message (plain + checksum) to process CHAN which simulates the transport network.



Process CHAN will replace the characters of the message with others randomly based on a probability (of change) which is provided externally as a parameter of the overall program. If the string-message on its way from P1 to P2 (and vice versa) is found at ENC? to be altered the checksum check will lead to rejection (of the incoming message). If the message, despite the random replacement process is received unchanged, it is decapsulated (from the control information) and provided to the upper-layer application (for visualization to the user). If the message (e.g., of P1), due to its intentional alteration, is received with a mismatch between string and checksum, process ENC2 will request the retransmission of the relevant information.

Sending the message TERM from a process P{1|2} and its passage through the rest (ENC?, CHAN, etc.) will signal the termination of all processes and the final release of the synchronization structures. The edges between the above processes will be implemented with shared memory segments (in/out structures) and will be protected by the appropriate semaphores.

Due Date: 2/Dec/2020
Submission method: upload to the appropriate folder of eclass. A tar file must be submitted containing all the relevant files: source and header files, makefile, etc. Accompanying material: limited documentation of 3-4 pages that explains the program and gives brief technical details. Implementation: the assignment is individual, it must be implemented in C/C++. The assignment will be examined on LINUX systems according to a schedule that will be announced after the due date.